

End-to-End ETL Data Pipeline and Data Warehouse for Multi-Source Integration at Pac-company



Description

This project demonstrates the development of an **end-to-end ETL (Extract, Transform, Load) data pipeline and data warehouse** for **Pac-company**, a new company that doesn't have a strong data architecture yet. The purpose is to combine and organize data from different sources into one analytical database that can be used for research and making decisions.

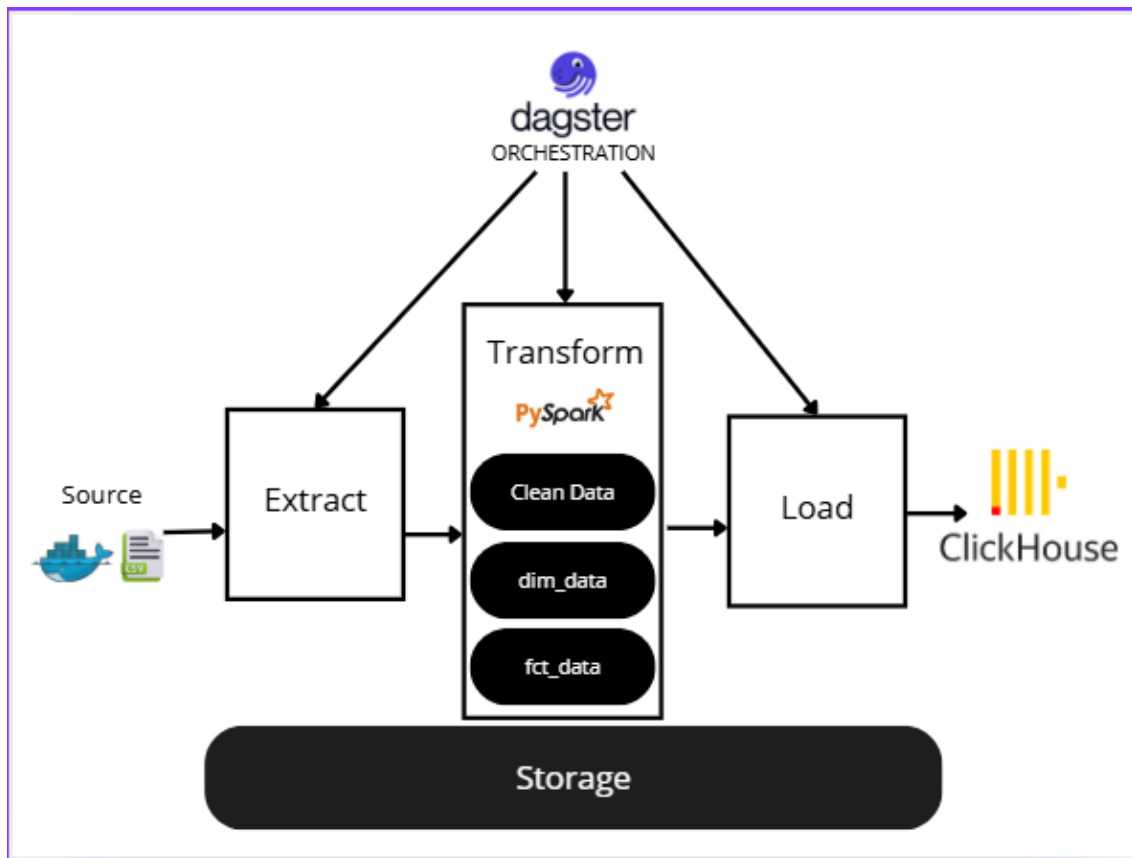
The pipeline consolidates three primary data sources:

- Sales data extracted from a PostgreSQL database comprising transactional records with incomplete and inconsistent entries.
- Product pricing information contained in a CSV file with unorganized and incomplete data.

Project Workflow

- **Requirements Gathering & Solution Design** – identifying stakeholder needs, data characteristics, and suitable tools
- **ETL Pipeline and Data Warehouse Design** – designing the data warehouse architecture following the Kimball dimensional modeling approach, defining fact and dimension tables, and illustrating the overall extraction, transformation, and loading flow.
- **Implementation using Dagster**– orchestrating modular ETL tasks with data transformation handled by **PySpark**.
- **Data Warehouse Construction in ClickHouse** – storing transformed datasets following **Kimball's dimensional modeling** for analytical efficiency.
- **Scheduling and Automation** – employing **logging** for periodic pipeline execution with detailed logging for monitoring and maintenance.

ETL Pipeline Design



Data Cleaning & Transformation Design

Sales Data design:

Column name	Change column name	Problem	Problem-solve
name	product_name	-	Change the column name
main_category	product_main_category	-	Change the column name
sub_category	product_sub_category	-	Change the column name
image	product_image_url	-	Change the column name
link	product_url	-	Change the column name
ratings	product_rating	there are null and not important value(such as FREE,Get,₹)	Change the column name, type data and impute null and another value to 0
no_of_ratings	product_rating_count	there are null and not important value(such as FREE,Usually dispatched,only)	Change the column name, type data and impute null value to 0
discount_price	product_price_discount	there are null value	Change the column name, create 10% discount of actual_price
actual_price	product_price	there are null value	Change the column name, drop null value
Unnamed: 0	-	Unknown column	Deleted
-	created_at	new timestamp column	

Product Data Design:

Column name	Change column name	Problem	Problem-solve
id	product_id	-	Change the column name,
amountMax	price_max	-	Change the column name
amountMin	price_min	-	Change the column name
availability	product_availability	there are null and not important value(TRUE,More on the way, Yes,yes,..)	impute data into available and not available
condition	product_condition	there are null and not important value(manufacturer refurbished, new, refurbished, pre-owned)	impute data into new, used and refurbished
currency	-	all data are USD	Deleted
dateSeen	seen_at	null data after cast type	drop null value
isSale	-	no need isSale when having a availability	deleted
merchant	product_merchant		Change the column name,
shipping	-	Drop	Deleted
sourceURLs	product_source	-	-
asins	-	-	-
brand	product_brand	-	Change the column name,
categories	product_categories	-	Change the column name,
dateadded	created_at	-	Change the column name,
dateupdated	updated_at	-	Change the column name,

ean	-	a lot of null value	deleted
imageURLs	product_image_url	-	Change the column name,
keys	product_key	-	Change the column name,
manufacturer	product_manufacturer	-	Change the column name,
name	product_name	-	Change the column name,
primary_categories	product_primary_categories	-	Change the column name,
sourceURLs.1	-	double column	deleted
upc	-	-	-
weight	product_weight	-	Change the column name,
unnamed:26-30	-	unknown column	deleted

Data Warehouse design

Process Business 1 – Sales Transaction Performance

Grain: one sales record per product.

- fct_sales_transaction
- dim_product

Business Process 2 (Product) – Product catalog and pricing standardization.

Grain: one product price record per product per merchant per date seen per product category.

- fct_product_price
- dim_product
- dim_merchant

- dim_date
- dim_product_category

Bus Matrix

Sales

fct table/dim_table	dim_product
fct_sales_transaction	✓

Product

fct table/dim_table	dim_product	dim_date	dim_merchant	dim_product_category
fct_product_price	✓	✓	✓	✓

Data Warehouse Schema

Sales

Dimension Table:

- dim_product
 - source: sales-clean.parquet
 - sink: Table_dim_product

coloumn_stg	column_dim	type_data pyspark	type_data warehouse
-	product_id[PK]	surrogate_key UUID	String
product_name	product_name	string	String
product_main_category	product_main_category	string	String
product_sub_category	product_sub_category	string	String
product_image_url	product_image_url	string	String
product_link	product_link	string	String

Fact Table:

- fct_sales_transaction:
 - source:
 - dim_table
 - sales_data

source	column_fct	relation	type_data pyspark	type_data warehouse
dim_product	sales_transaction_id [PK]		surrogate_key UUID	String
sales-clean.parquet	product_id [FK]	dim_product	UUID	String
sales-clean.parquet	product_rating		float	FLOAT32
sales-clean.parquet	product_rating_count		int	INT32
sales-clean.parquet	product_discount_price		float	FLOAT32
sales-clean.parquet	product_price		int	INT32

Product

Dimension Table:

- dim_product
 - source: stg_sales_clean.parquet
 - sink: dim_product

coloumn_raw	column_dim	type_data pyspark	type_data warehouse
-	sk_product_id [PK]	surrogate_key UUID	String
product_id	nk_product_id	string	String
product_name	product_name	string	String
product_image_url	product_image_url	string	String
product_source	product_source	string	String
upc	product_upc	string	String
asins	product_asins	string	String
product_key	product_key	string	String
product_availability	product_availability	string	String
generate when load to clickhouse	dataAdded	-	DateTime DEFAULT now()
generate when load to clickhouse	dateUpdate		DateTime DEFAULT now(

- dim_date
 - source: generate time
 - sink: dim_product

column_dim	type_data pyspark	type_data warehouse
date_id [PK]	YYYYMMDD int	Int32
full_date	Date	DATE
day	int	Int32
month	int	Int8
year	int	Int32

- dim_merchant
 - source: product_clean.parquet
 - sink: dim_data

coloumn_raw	column_dim	type_data pyspark	type_data warehouse
-	merchant_id [PK]	surrogate_key UUID	String
product_merchant	merchant_name	string	String

- dim_product_category
 - source: product_clean.parquet
 - sink: Table_dim_product

coloumn_raw	column_dim	type_data pyspark	type_data warehouse
-	product_catetogry_id [PK]	surrogate_key UUID	String
product_primary_categories	product_primary_categories	string	String
product_categories	product_categories	string	String

Fact Table:

- fct_product_price:
 - source:
 - dim_product
 - dim_date
 - dim_merchant
 - dim_product_category
 - product_data

source	column_fct	relation	type_data pyspark	type_data warehouse
-	product_price_id [PK]		surrogate_key UUID	String
dim_product	sk_product_id [FK]	dim_product	String	String
dim_date	date_id [FK]	dim_date	int	Int32
product_clean.pa rquet	seen_at		timestamp	DATETIME
dim_merchant	merchant_id [FK]	dim_mercha nt	string	String
dim_product_cate gory	product_category_id [FK]	dim_product _category	int	String
product_clean.pa rquet	price_max		int	int32
product_clean.pa rquet	price_min		int	int32
dim_product	dateAdded		timestamp	DATETIME
dim_product	dateUpdated		timestamp	DATETIME